

Optimization-based machine learning for computational imaging

Proximal algorithms

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Proximal algorithms

→ Minimisation problem :

$$\hat{x} \in \underset{x}{\operatorname{Argmin}} f_1(x) + f_2(x)$$

with f_1 and f_2 either diff. with Lipschitz gradient or proximable.

→ Design of a recursive sequence of the form:

$$(\forall k \in \mathbb{N}) \quad x^{[k+1]} = \mathbf{T}x^{[k]},$$

Gradient descent $\mathbf{T} = \operatorname{Id} - \tau(\nabla f_1 + \nabla f_2)$

Proximal point $\mathbf{T} = \operatorname{prox}_{\tau(f_1+f_2)}$

Forward-Backward $\mathbf{T} = \operatorname{prox}_{\tau f_2}(\operatorname{Id} - \tau \nabla f_1)$

Peaceman-Rachford $\mathbf{T} = (2\operatorname{prox}_{\tau f_2} - \operatorname{Id}) \circ (2\operatorname{prox}_{\tau f_1} - \operatorname{Id})$

Douglas-Rachford $\mathbf{T} = \operatorname{prox}_{\tau f_2}(2\operatorname{prox}_{\tau f_1} - \operatorname{Id}) + \operatorname{Id} - \operatorname{prox}_{\tau f_1}$

Fixed point algorithm: zeros and fixed points

Let $\mathcal{H}$ be a Hilbert space. Let $\Phi: \mathcal{H} \rightarrow 2^{\mathcal{H}}$ and $\mathbf{T}: \mathcal{H} \rightarrow 2^{\mathcal{H}}$.

The set of **fixed points** of $\mathbf{T}$ is : $\text{Fix}\mathbf{T} = \{x \in \mathcal{H} \mid x \in \mathbf{T}x\}$.

The set of **zeros** of Φ is : $\text{zer}\Phi = \{x \in \mathcal{H} \mid 0 \in \Phi x\}$.

Minimisation problem and Fermat rule

$$\hat{x} \in \underset{x \in \mathbb{R}^N}{\text{Argmin}} f(x) \Leftrightarrow \nabla f(\hat{x}) = 0 \Leftrightarrow \hat{x} \in \text{zer}\nabla f$$

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Minimisation problem and Fermat rule

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Fix point and zeros for gradient descent:

$$\hat{x} \in \text{Fix}(\text{Id} - \nabla f) \Leftrightarrow \hat{x} = (\text{Id} - \nabla f)\hat{x} \Leftrightarrow \hat{x} \in \text{zer}\nabla f$$

Remark: $\mathbf{T} = \text{Id} - \nabla f$ and $\Phi = \nabla f$

Fixed point algorithm: convergence

Let $\mathcal{H}$ be a Hilbert space, $(x^{[k]})_{k \in \mathbb{N}}$ be a sequence in $\mathcal{H}$ and $\hat{x} \in \mathcal{H}$.

- $(x^{[k]})_{k \in \mathbb{N}}$ **converges strongly** to $\hat{x}$ if

$$\lim_{k \rightarrow \infty} \|x^{[k]} - \hat{x}\| = 0.$$

It is denoted by $x^{[k]} \rightarrow \hat{x}$.

- $(x^{[k]})_{k \in \mathbb{N}}$ **converges weakly** to $\hat{x}$ if

$$(\forall u \in \mathcal{H}) \quad \lim_{n \rightarrow \infty} \langle u, x^{[k]} - \hat{x} \rangle = 0.$$

It is denoted by $x^{[k]} \rightharpoonup \hat{x}$.

Remark: In a finite dimensional Hilbert space, strong and weak convergences are equivalent.

Banach-Picard theorem

$\mathbf{T}: \mathcal{H} \rightarrow \mathcal{H}$ is ω –**Lipschitz continuous** for some $\omega > 0$ if

$$(\forall \mathbf{x} \in \mathcal{H})(\forall \mathbf{u} \in \mathcal{H}) \quad \|\mathbf{T}\mathbf{x} - \mathbf{T}\mathbf{u}\| \leq \omega \|\mathbf{x} - \mathbf{u}\|.$$

$\mathbf{T}$ is **nonexpansive** if it is 1–Lipschitz continuous.

Banach-Picard theorem:

Let $\omega \in [0, 1[$, $\mathbf{T}: \mathcal{H} \rightarrow \mathcal{H}$ be a ω –Lipschitz continuous operator, and $\mathbf{x}^{[0]} \in \mathcal{H}$.

Set

$$(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \mathbf{T}\mathbf{x}^{[k]}.$$

Then, $\text{Fix}\mathbf{T} = \{\hat{\mathbf{x}}\}$ for some $\hat{\mathbf{x}} \in \mathcal{H}$ and we have

$$(\forall k \in \mathbb{N}) \quad \|\mathbf{x}^{[k]} - \hat{\mathbf{x}}\| \leq \omega^k \|\mathbf{x}_0 - \hat{\mathbf{x}}\|.$$

Moreover, $(\mathbf{x}^{[k]})_{k \in \mathbb{N}}$ converges strongly to $\hat{\mathbf{x}}$ with linear convergence rate ω .

Averaged nonexpansive operator

An operator $\mathbf{T}: \mathcal{H} \rightarrow \mathcal{H}$ is μ -averaged nonexpansive for some $\mu \in]0, 1]$ if, for every $x \in \mathcal{H}$ and $u \in \mathcal{H}$,

$$\|\mathbf{T}x - \mathbf{T}u\|^2 \leq \|x - u\|^2 - \left(\frac{1 - \mu}{\mu}\right) \|(\text{Id} - \mathbf{T})x - (\text{Id} - \mathbf{T})u\|^2,$$

$\mathbf{T}$ is firmly nonexpansive if it is $1/2$ -averaged.

$\mathbf{T}$ is nonexpansive if and only if $\mathbf{T}$ is 1 -averaged.

Theorem:

Let $\mu \in]0, 1[$, let $\mathbf{T}: \mathcal{H} \rightarrow \mathcal{H}$ be a μ -averaged nonexpansive operator such that $\text{Fix}\mathbf{T} \neq \emptyset$, and let $x^{[0]} \in \mathcal{H}$. Set

$$(\forall k \in \mathbb{N}) \quad x^{[k+1]} = \mathbf{T}x^{[k]}.$$

Then $(x^{[k]})_{k \in \mathbb{N}}$ converges weakly to a point in $\text{Fix}\mathbf{T}$.

Nonlinear operators

Properties of f	$\mathbf{T}$	ω -Lipschitz	μ -averaged
f convex ∇f β -Lipschitz	$\text{Id} - \tau \nabla f$ $\tau \in (0, 2\beta^{-1})$	$\omega = 1$	$\mu = \frac{\tau\beta}{2}$
f ρ -strongly convex ∇f β -Lipschitz	$\text{Id} - \tau \nabla f$ $\tau \in (0, 2\beta^{-1})$	$\omega = \max\{(1 - \tau\rho), (\tau\beta - 1)\}$	$\mu = \frac{1+\omega}{2}$
$f \in \Gamma_0$	$\text{prox}_{\tau f}$ $\tau > 0$	$\omega = 1$	$\mu = \frac{1}{2}$
f ρ -strongly convex	$\text{prox}_{\tau f}$ $\tau > 0$	$\omega = (1 + \tau\rho)^{-1}$	$\mu = \frac{1+\omega}{2}$

[Taylor, J. M. Hendrickx, and F. Glineur] [Bauschke, Combettes, 2017]

Strong convexity

Let $f \in \Gamma_0(\mathcal{H})$. f is ρ -strongly convex with $\rho > 0$ if $f - \frac{\rho}{2}\|\cdot\|_2^2$ is convex.

Properties:

- If f is ρ -strongly convex then

$$(\forall x, y \in \mathcal{H}) \quad \langle \nabla f(x) - \nabla f(y) | x - y \rangle \geq \rho \|x - y\|^2$$

- If f is twice differentiable, then f is ρ -strongly convex if and only if all the eigenvalues of the Hessian matrix of f are at most equal to ρ .

Let $f \in \Gamma_0(\mathbb{R}^N)$ differentiable with a β -Lipschitz gradient. We set, for some $\tau > 0$,

$$\mathbf{T} := \operatorname{Id} - \tau \nabla f$$

- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \mathbf{x}^{[k]} - \tau \nabla f(\mathbf{x}^{[k]})$.

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- For every $\tau > 0$, $\operatorname{zer} \nabla f = \operatorname{Fix} \mathbf{T}$.

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- For every $\tau > 0$, $\operatorname{zer} \nabla f = \operatorname{Fix} \mathbf{T}$.
- $\mathbf{T} = \operatorname{Id} - \tau \nabla f$ is a $\tau\beta/2$ -averaged operator.
→ cf. Proposition 4.39 in [Bauschke-Combettes, 2017]

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- $\mathbf{T} = \operatorname{Id} - \tau \nabla f$ is a $\tau\beta/2$ -averaged operator.
 → cf. Proposition 4.39 in [Bauschke-Combettes, 2017]
- **Convergence:** For every $\tau \in]0, 2\beta^{-1}[$, the gradient method **converges** to a point in $\operatorname{zer} \nabla f$.

Let $f \in \Gamma_0(\mathbb{R}^N)$. We set, for some $\tau > 0$,

$$\mathbf{T} := \operatorname{prox}_{\tau f}$$

- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \operatorname{prox}_{\gamma f}(\mathbf{x}^{[k]}).$

Let $f \in \Gamma_0(\mathbb{R}^N)$. We set, for some $\tau > 0$,

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- **Iterations:** $(\forall k \in \mathbb{N}) \quad x^{[k+1]} = \operatorname{prox}_{\gamma f}(x^{[k]}).$
- For every $\tau > 0$, $\operatorname{Fix} \mathbf{T} = \operatorname{zer} \partial f$.

Proof:

$$\begin{aligned} x = \operatorname{prox}_{\tau f} x &\Leftrightarrow x \in (\mathbf{I} + \tau \partial f)x \\ &\Leftrightarrow x \in x + \tau \partial f(x) \\ &\Leftrightarrow 0 \in \partial f \end{aligned}$$

Let $f \in \Gamma_0(\mathbb{R}^N)$. We set, for some $\tau > 0$,

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- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \operatorname{prox}_{\gamma f}(\mathbf{x}^{[k]})$.
- For every $\tau > 0$, $\operatorname{Fix} \mathbf{T} = \operatorname{zer} \partial f$.
- For every $\tau > 0$ and any $f \in \Gamma_0(\mathcal{H})$, $\operatorname{prox}_{\tau f}$ is 1/2-averaged.
→ cf. s.10 or Proposition 23.8 in [Bauschke-Combettes, 2017]

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- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \operatorname{prox}_{\gamma f}(\mathbf{x}^{[k]})$.
- For every $\tau > 0$, $\operatorname{Fix} \mathbf{T} = \operatorname{zer} \partial f$.
- For every $\tau > 0$ and any $f \in \Gamma_0(\mathcal{H})$, $\operatorname{prox}_{\tau f}$ is $1/2$ -averaged.
→ cf. s.10 or Proposition 23.8 in [Bauschke-Combettes, 2017]
- The **PPA method converges** to a point in $\operatorname{zer} \partial f$.

Objective Let $f_1: \mathbb{R}^N \rightarrow \mathbb{R}$ a convex, proper and β -Lipschitz differentiable function and $f_2 \in \Gamma_0(\mathbb{R}^N)$. We set, for some $\tau > 0$,

$$\mathbf{T} := \operatorname{prox}_{\tau f_2} \circ (\operatorname{Id} - \tau \nabla f_1)$$

- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \operatorname{prox}_{\tau f_2}(\mathbf{x}^{[k]} - \tau \nabla f_1(\mathbf{x}^{[k]})).$

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- **Iterations:** $(\forall k \in \mathbb{N}) \quad x^{[k+1]} = \operatorname{prox}_{\tau f_2}(x^{[k]} - \tau \nabla f_1(x^{[k]}))$.
- Roots in projected gradient method [Levitin 1966] when $g = \iota_C$ for some closed convex set C .

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- **Iterations:** $(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \operatorname{prox}_{\tau f_2}(\mathbf{x}^{[k]} - \tau \nabla f_1(\mathbf{x}^{[k]}))$.
- For every $\tau > 0$, $\operatorname{zer}(\nabla f_1 + \partial g) = \operatorname{Fix} \mathbf{T}$.

Proof:

$$\begin{aligned} \mathbf{x} \in \operatorname{Fix} \mathbf{T} &\Leftrightarrow (\operatorname{Id} - \tau \nabla f_1) \mathbf{x} \in (\operatorname{Id} + \tau \partial f_2) \mathbf{x} \\ &\Leftrightarrow 0 \in \nabla f_1(\mathbf{x}) + \partial f_2(\mathbf{x}). \end{aligned}$$

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- $\operatorname{prox}_{\tau f_2}(\operatorname{Id} - \tau \nabla f_1)$ is μ -averaged nonexpansive where $\mu = \frac{\mu_1 + \mu_2 - 2\mu_1\mu_2}{1 - \mu_1\mu_2}$ with $\mu_2 = \tau\beta/2$ and $\mu_1 = 1/2$ leading to $\mu = \frac{1}{2 - \tau\beta/2} \in]0, 1[$ and $\tau < 2/\beta$.

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- For every $\tau \in]0, 2/\beta[$, the **FBS converges** to a point in $\operatorname{zer}(\nabla f_1 + \partial f_2)$.

$$\hat{\mathbf{x}} \in \operatorname{Argmin}_{\mathbf{x} \in \mathbb{R}^N} \left\{ f(\mathbf{x}) = f_1(\mathbf{x}) + f_2(\mathbf{x}) \right\}$$

Objective Let $f_1: \mathbb{R}^N \rightarrow \mathbb{R}$ a convex, proper and β -Lipschitz differentiable function and $f_2 \in \Gamma_0(\mathbb{R}^N)$. We set, for some $\tau > 0$,

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Theorem [Combettes & Wajs, 2005]: Let $(x^{[k]})_{k \in \mathbb{N}}$ be a sequence generated by the FB algorithm. Let $0 < \tau < 2\beta^{-1}$. Then

- $(x^{[k]})_{k \in \mathbb{N}}$ converges to a minimiser $\hat{x}$ of f
- $(f(x^{[k]}))_{k \in \mathbb{N}}$ is a non-increasing sequence converging to $f(\hat{x})$.

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Theorem [Briceno-Arias & Pustelnik, 2005]:

If additionally, f_1 is ρ -strongly convex. Suppose that $\tau \in]0, 2\beta^{-1}[$. Then $\mathbf{T}$ is $\omega(\tau)$ -Lipschitz continuous with

$$\omega(\tau) := \max \{ |1 - \tau\rho|, |1 - \tau\beta| \} \in]0, 1[.$$

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$$\omega(\tau) := \max \{ |1 - \tau\rho|, |1 - \tau\beta| \} \in]0, 1[.$$

In particular, the minimum is achieved at

$$\tau^* = \frac{2}{\rho + \beta} \quad \text{and} \quad \omega(\tau^*) = \frac{\beta - \rho}{\beta + \rho}.$$

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Theorem [Briceno-Arias & Pustelnik, 2023][Briceno-Arias, 2025]:

If additionally, f_2 is ρ -strongly convex. Suppose that $\tau \in]0, 2\beta^{-1}]$. Then $\mathbf{T}$ is $\omega(\tau)$ -Lipschitz continuous with

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In particular, the minimum is achieved at

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- Convergence may be slow in practice...
 - 👉 Use Nesterov acceleration (*inertia*)
 - 👉 Use second order information (*preconditioning*)
 - 👉 Use multilevel strategy
- What if $\operatorname{prox}_{\gamma_k g}$ does not have a closed form?
 - 👉 Use sub-iterations (e.g. dual FB algorithm)
 - 👉 Use more advanced methods (e.g. primal-dual algorithms)

What is inertia?

Goal: Inertia aims to use information from the **previous iterate(s)** $(\mathbf{x}^{[k']})_{k' \leq k}$ to build the next iterate $\mathbf{x}^{[k+1]}$.

Why? Use memory to go faster!

For FB we have:

$$(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \mathbf{T}_k(\mathbf{x}^{[k]}) \text{ where } \mathbf{T}_k = \text{prox}_{\tau f_2} \circ (\text{Id} - \tau \nabla f_1)$$

Introducing inertia would lead to

$$(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \tilde{\mathbf{T}}_k(\mathbf{x}^{[1]}, \dots, \mathbf{x}^{[k]})$$

QUESTION: How to choose $\tilde{\mathbf{T}}_k$?

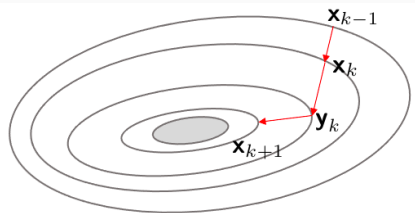
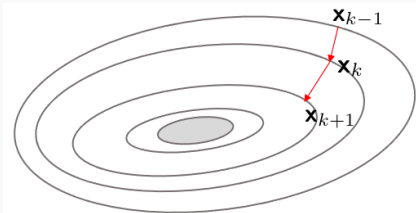
Particular case: Inertia for GD algorithm

- Let $f_2 \equiv 0$. In this case $\text{prox}_{f_2} = \text{Id}$.
- The *path* taken by the iterates $(\mathbf{x}^{[k]})_{k \in \mathbb{N}}$ is determined by the opposite of the gradient direction:

$$(\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} = \mathbf{x}^{[k]} - \tau_k \nabla f_1(\mathbf{x}^{[k]})$$

- **Acceleration:** *Nesterov-type accelerated GD algorithm* [Nesterov, 83]

$$\begin{aligned} (\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} &= \mathbf{y}^{[k]} - \tau \nabla f_1(\mathbf{y}^{[k]}) \quad \text{with } \tau \in]0, 1/\beta[\\ \mathbf{y}^{[k+1]} &= \mathbf{x}^{[k+1]} + \alpha_k (\mathbf{x}^{[k+1]} - \mathbf{x}^{[k]}) \end{aligned}$$



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$$\begin{aligned} (\forall k \in \mathbb{N}) \quad \mathbf{x}^{[k+1]} &= \mathbf{y}^{[k]} - \tau \nabla f_1(\mathbf{y}^{[k]}) \quad \text{with } \tau \in]0, 1/\beta] \\ \mathbf{y}^{[k+1]} &= \mathbf{x}^{[k+1]} + \alpha_k (\mathbf{x}^{[k+1]} - \mathbf{x}^{[k]}) \end{aligned}$$

- Each iteration takes nearly the same computational cost as GD
- **not** a *descent* method (i.e. we may not have $f_1(\mathbf{x}^{[k+1]}) \leq f_1(\mathbf{x}^{[k]})$)

Inertial FB

For $k = 0, 1, \dots$

$$\left[\begin{array}{l} \text{Let } \gamma_k \in]0, 1/\beta] \\ x^{[k+1]} = \text{prox}_{\tau_k f_2} \left(y^{[k]} - \tau_k \nabla f_1(y^{[k]}) \right) \\ y^{[k+1]} = x^{[k+1]} + \alpha_k (x^{[k+1]} - x^{[k]}) \end{array} \right.$$

- [Beck and Teboulle, 2009]

Adopt the inertia (momentum) strategy proposed by Nesterov

$$\alpha_k = \frac{\theta_k - 1}{\theta_{k+1}} \quad \text{with} \quad \theta_{k+1} = \frac{1 + \sqrt{1 + 4\theta_k^2}}{2}$$

- [Chambolle and Dossal, 2015]

Different rule

$$\alpha_k = \frac{\theta_k - 1}{\theta_{k+1}} \quad \text{with} \quad \theta_{k+1} = \left(\frac{k+a}{a} \right)^d$$

with $d \in]0, 1]$ and $a > \max\{1, (2d)^{1/d}\}$.

Convergence of Inertial FB

Let $(x^{[k]})_{k \in \mathbb{N}}$ be generated by **FB iterations** with $\tau \in]0, \beta^{-1}[$.

$(f(x^{[k]}))_{k \in \mathbb{N}}$ converges to $f(\hat{x})$ at the rate $O(1/k)$:

$$f(x^{[k]}) - f(\hat{x}) \leq \frac{\beta}{2k} \|x^{[0]} - \hat{x}\|^2$$

Let $(x^{[k]})_{k \in \mathbb{N}}$ be generated by **Inertial FB**.

$(f(x^{[k]}))_{k \in \mathbb{N}}$ converges to $f(\hat{x})$ at the rate $O(1/k^2)$:

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Convergence of Inertial FB

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Proof: A complete proof is provided in [\[Beck and Teboulle, 2009\]](#).

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- Improved iteration complexity:
 - FB: $f(x^{[k]}) - f(\hat{x}) \approx O(1/k)$
 - FISTA: $f(x^{[k]}) - f(\hat{x}) \approx O(1/k^2)$
- (Almost) Same computational complexity per iteration as FB
- **Issue**: Convergence guaranties of the sequence $(x^{[k]})_{k \in \mathbb{N}}$?

Convergence of Inertial FB

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Let $(x^{[k]})_{k \in \mathbb{N}}$ be generated by **Inertial FB** with Chambolle-Dossal rule $\alpha_k = \frac{\theta_k - 1}{\theta_{k+1}}$ with $\theta_{k+1} = \left(\frac{k+a}{a}\right)^d$ with $d \in]0, 1]$ and $a > \max\{1, (2d)^{1/d}\}$

Then the sequence $(x^{[k]})_{k \in \mathbb{N}}$ converges to a minimiser of f .

Duality

Minimization problem

Find

$$\hat{x} \in \operatorname{Argmin}_{x \in \mathbb{R}^N} f_1(x) + f_2(x) + g(Dx)$$

- $f_1: \mathbb{R}^N \rightarrow \mathbb{R}$ is convex and β -Lipschitz differentiable
- $f_2 \in \Gamma_0(\mathbb{R}^N)$
- $g \in \Gamma_0(\mathbb{R}^M)$ and $D \in \mathbb{R}^{M \times N}$

Use FB algorithm ?

For $k = 0, 1, \dots$

$$\left\{ \begin{array}{l} x^{[k+1]} = \operatorname{prox}_{\tau(f_2+g \circ D)}(x^{[k]} - \tau \nabla f_1(x^{[k]})) \end{array} \right.$$

➤ How to compute $\operatorname{prox}_{\tau(f_2+g \circ D)}$?

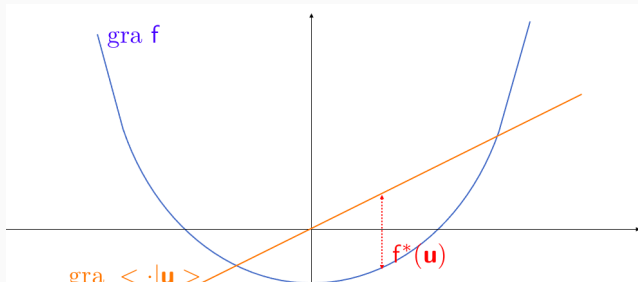
➤ Use primal-dual methods

Conjugate function

Conjugate of a function $f: \mathcal{H} \rightarrow]-\infty, +\infty]$:

$$\begin{aligned} f^*: \quad \mathcal{H} &\rightarrow [-\infty, +\infty] \\ u &\mapsto \sup_{x \in \mathbb{R}^N} \langle x \mid u \rangle - f(x) \end{aligned}$$

Graphical illustration: $f^*(u)$ is the supremum of the signed vertical distance between the graph of f and that of the continuous linear functional $\cdot u$

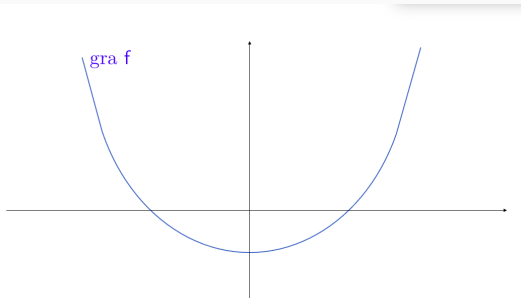


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Graphical illustration: Second interpretation - How to build $f^*(u)$?

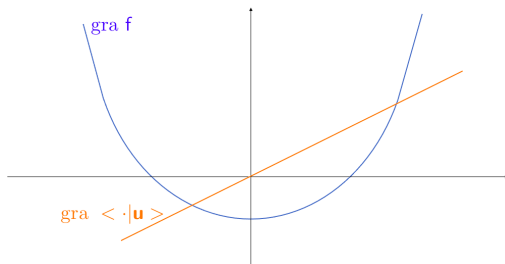


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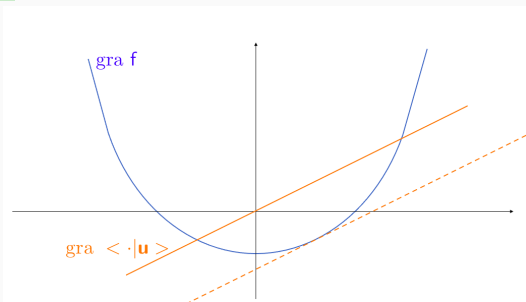


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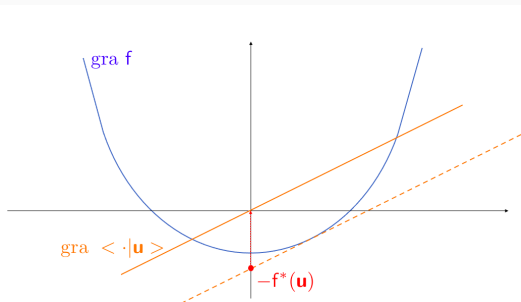


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Examples:

f	f^*
$\frac{1}{2} \ \cdot \ ^2$	$\frac{1}{2} \ \cdot \ ^2$
$\lambda \ \cdot \ _1$	$\iota_{\mathcal{B}_\infty(0, \lambda)}$ with $\lambda > 0$
$\frac{1}{q} \ \cdot \ _q^q$	$\frac{1}{q^*} \ \cdot \ _{q^*}^{q^*}$ with $1/q + 1/q^* = 1$

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Moreau-Fenchel theorem

$f: \mathcal{H} \rightarrow]-\infty, +\infty]$ be a proper function.

$$f \text{ is l.s.c. and convex} \Leftrightarrow f^{**} = f.$$

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Moreau decomposition formula

Let $\mathcal{H}$ be a Hilbert space, $f \in \Gamma_0(\mathcal{H})$ and $\gamma > 0$.

$$(\forall x \in \mathcal{H}) \quad \text{prox}_{\gamma f^*} x = x - \gamma \text{prox}_{\gamma^{-1} f}(\gamma^{-1} x).$$

Dual methods

Fenchel-Rockafellar duality

Primal problem

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$, $g: \mathcal{G} \rightarrow]-\infty, +\infty]$. Let $D \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.

We want to

$$\underset{x \in \mathcal{H}}{\text{minimize}} \quad f(x) + g(Dx).$$

Dual problem

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$, $g: \mathcal{G} \rightarrow]-\infty, +\infty]$. Let $D \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.

We want to

$$\underset{u \in \mathcal{G}}{\text{minimize}} \quad f^*(-D^*u) + g^*(u).$$

Fenchel-Rockafellar duality

Weak duality

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

Let f be a proper function from $\mathcal{H}$ to $] - \infty, +\infty]$, g be a proper function from $\mathcal{G}$ to $] - \infty, +\infty]$, and $D \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Let

$$\mu = \inf_{x \in \mathcal{H}} f(x) + g(Dx) \quad \text{and} \quad \mu^* = \inf_{u \in \mathcal{G}} f^*(-D^*u) + g^*(u).$$

We have $\mu \geq -\mu^*$. If $\mu \in \mathbb{R}$, $\mu + \mu^*$ is called the **duality gap**.

Fenchel-Rockafellar duality

Weak duality

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We have $\mu \geq -\mu^*$. If $\mu \in \mathbb{R}$, $\mu + \mu^*$ is called the **duality gap**.

Proof: According to Fenchel-Young inequality,

$$f(x) + g(Dx) + f^*(-D^*u) + g^*(u) \geq \langle x \mid -D^*u \rangle + \langle Dx \mid u \rangle = 0.$$

Strong duality

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{G})$, and $D \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.

If $(\text{dom} g) \cap D(\text{dom} f) \neq \emptyset$ or $\text{dom} g \cap (D(\text{dom} f)) \neq \emptyset$, then

$$\mu = \inf_{x \in \mathcal{H}} f(x) + g(Dx) = - \min_{u \in \mathcal{G}} f^*(-D^*u) + g^*(u) = -\mu^* .$$

Duality theorem

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{G})$, and $D \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.

- If there exists $\hat{x} \in \mathcal{H}$ such that $0 \in \partial f(\hat{x}) + D^* \partial g(D\hat{x})$, then $\hat{x}$ is a solution to the primal problem. Moreover, there exists a solution $\hat{u}$ to the dual problem such that $-D^* \hat{u} \in \partial f(\hat{x})$ and $D\hat{x} \in \partial g^*(\hat{u})$.
- If there exists $(\hat{x}, \hat{u}) \in \mathcal{H} \times \mathcal{G}$ such that $-D^* \hat{u} \in \partial f(\hat{x})$ and $D\hat{x} \in \partial g^*(\hat{u})$ then $\hat{x}$ (resp. $\hat{u}$) is a solution to the primal (resp. dual) problem.

If $(\hat{x}, \hat{u}) \in \mathcal{H} \times \mathcal{G}$ is such that $-D^* \hat{u} \in \partial f(\hat{x})$ and $D\hat{x} \in \partial g^*(\hat{u})$, then $(\hat{x}, \hat{u})$ is called a **Kuhn-Tucker point**.

Fenchel-Rockafellar duality

Particular case :

If $f = \varphi + \frac{1}{2} \|\cdot - z\|^2$ where $\varphi \in \Gamma_0(\mathcal{H})$ and $z \in \mathcal{H}$, then

$$\begin{aligned} -D^*\hat{u} \in \partial f(\hat{x}) &\Leftrightarrow -D^*\hat{u} \in \partial\varphi(\hat{x}) + \hat{x} - z \\ &\Leftrightarrow 0 \in \hat{x} + D^*\hat{u} - z + \partial\varphi(\hat{x}). \end{aligned}$$

Hence,

$$\hat{x} = \text{prox}_{\varphi}(-D^*\hat{u} + z).$$

Dual FB algorithm

Let $z \in \mathbb{R}^N$, $g \in \Gamma_0(\mathbb{R}^M)$ and $D \in \mathbb{R}^{M \times N}$.

Primal problem: $\hat{x} = \arg \min_{x \in \mathbb{R}^N} \frac{1}{2} \|x - z\|^2 + g(Dx)$

Dual problem: $\hat{u} \in \operatorname{Argmin}_{u \in \mathbb{R}^M} \frac{1}{2} \|z - D^*u\|^2 + g^*(u)$

Choose $u^{[0]} \in \mathbb{R}^M$ and $\tau \in]0, 2/\|D\|^2[$.

For $k = 0, 1, \dots$

$$\begin{cases} x^{[k]} = z - D^*u^{[k]} \\ u^{[k+1]} = \operatorname{prox}_{\tau g^*}(u^{[k]} + \tau Dx^{[k]}) \end{cases}$$

[Combettes, Dung, Vũ, 2011]

- The sequence $(u^{[k]})_{k \in \mathbb{N}}$ converges to a solution to the dual problem $\hat{u}$.
- The sequence $(x^{[k]})_{k \in \mathbb{N}}$ converges to a solution to the primal problem

$$\hat{x} = z - D^*\hat{u}.$$

Chambolle-Pock algorithm and strong convexity $\hat{x} \in \operatorname{Argmin}_{x \in \mathbb{R}^N} f_1(x) + g(Dx)$

CHAMBOLLE-POCK ALGORITHM: [Chambolle & Pock, 2011]

PROBLEM: Choose $\tau > 0$ and $\sigma > 0$ such that $\sigma\tau\|D\|^2 < 1$.

For $k = 0, 1, \dots$

$$\begin{cases} x^{[k+1]} = \operatorname{prox}_{\tau f_1}(x^{[k]} - \tau D^* u^{[k]}) \\ u^{[k+1]} = \operatorname{prox}_{\sigma g^*}(u^{[k]} + \sigma D(2x^{[k+1]} - x^{[k]})) \end{cases}$$

ACCELERATED VERSION: f_1 ρ -strongly convex [Chambolle & Pock, 2011]

Choose $\tau_0 > 0$ and $\sigma_0 > 0$ such that $\sigma_0\tau_0\|D\|^2 < 1$.

For $k = 0, 1, \dots$

$$\begin{cases} x^{[k+1]} = \operatorname{prox}_{\tau_k f_1}(x^{[k]} - \tau_k D^* u^{[k]}) \\ \alpha_k = (1 + 2\rho\tau_k)^{-1/2} \\ \tau_{k+1} = \alpha_k \tau_k \\ \sigma_k = \sigma_k \alpha_k^{-1/2} \\ y^{[k+1]} = x^{[k+1]} + \alpha_k(x^{[k+1]} - x^{[k]}) \\ u^{[k+1]} = \operatorname{prox}_{\sigma_{k+1} g^*}(u^{[k]} + \sigma_k D y^{[k+1]}) \end{cases}$$

Optimization algorithms

Forward-Backward	$f_1 + f_2$	f_1 grad. Lipschitz prox_{f_2}	[Combettes,Wajs,2005]
ISTA	$f_1 + f_2$	f_1 grad. Lipschitz $f_2 = \lambda \ \cdot\ _1$	[Daubechies et al, 2003]
Douglas-Rachford	$f_1 + f_2$	prox_{f_1} prox_{f_2}	[Combettes,Pesquet, 2007]
PPXA	$\sum_i f_i$	prox_{f_i}	[Combettes,Pesquet, 2008]
PPXA+	$\sum_i g_i \circ D_i$	prox_{g_i} $(\sum_{i=1}^m D_i^* D_i)^{-1}$	[Pesquet, Pustelnik, 2012]
ADMM	$f + g \circ D$	prox_f $(D^* D)^{-1}$	[Eckstein, Yao, 2015]
Chambolle-Pock	$f + g \circ D$	prox_f prox_g	[Chambolle, Pock, 2011]
Condat-Vũ	$f_1 + f_2 + g \circ D$	prox_f prox_g f_2 grad. Lipschitz	[Condat, 2013][Vũ, 2013]

Inference framework: feed-forward NN

$$\begin{aligned}(\forall \mathbf{x}^{[0]} \in \mathbb{R}^{N_0}) \quad \mathbf{x}^{[K]} &= \mathbf{f}_{\Theta}^K(\mathbf{x}^{[0]}) \\ &= \mathfrak{T}_{\Theta_K} \circ \dots \circ \mathfrak{T}_{\Theta_1}(\mathbf{x}^{[0]}),\end{aligned}$$

Layer/iteration

$$\mathfrak{T}_{\Theta_k} : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k} : \mathbf{x} \mapsto \mathfrak{D}_k(\mathbf{L}_k \cdot + \mathbf{b}_k),$$

- $\mathbf{L}_k : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k}$: linear operator,
- $\mathbf{b}_k \in \mathbb{R}^{N_k}$: shift parameter,
- $\mathfrak{D}_k : \mathbb{R}^{N_k} \rightarrow \mathbb{R}^{N_k}$: nonlinear operator parametrized by Λ_k .

Parameters: $\Theta = \cup_{k=1}^K \Theta_k$ with $\Theta_k = \{\Lambda_k, \mathbf{L}_k, \mathbf{b}_k\}$.

Full picture

